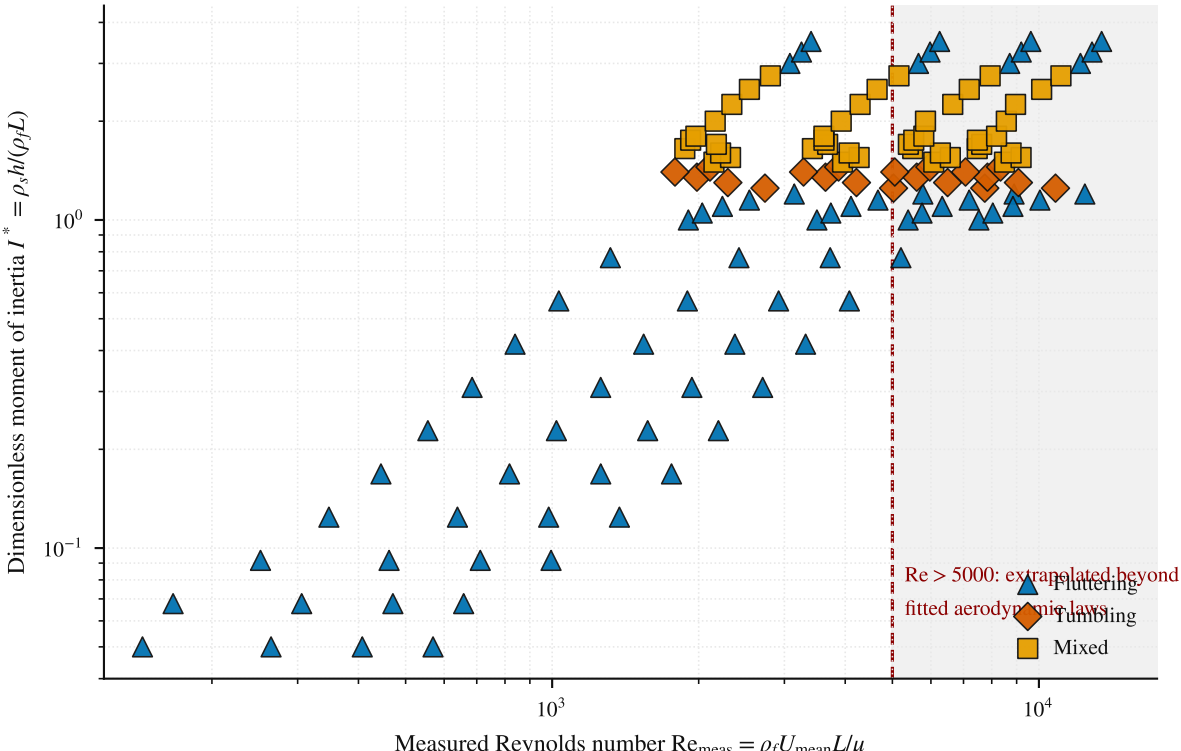
Falling regime map in ($\mathrm{Re}_{\mathrm{meas}}$, I^*) space ($L = 8, 12, 16, 20$ mm)
Dimensionless moment of inertia $I^* = \rho_s h l/(\rho_f L)$
10^0
10^{-1}
10^3
10^4
Measured Reynolds number $\mathrm{Re}_{\mathrm{meas}} = \rho_f U_{\mathrm{mean}} L/\mu$
Re > 5000: extrapolated beyond
fitted aerodynamic laws
Fluttering
Tumbling
Mixed